

Doudou Cao

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RESEARCH PROFILE

I am a biological anthropologist and osteoarchaeologist studying how ancient populations experienced health, growth, violence, and adaptation across environmental and historical change. My research uses human skeletal and morphometric data to model long-term historical change and population processes in China, with particular emphasis on the Tibetan Plateau and mountainous southwest China. I combine osteological analysis, 3D digital methods, spatial analysis, and Bayesian spatiotemporal modelling to examine interactions among biological variation, environmental pressure, and social organisation.

RESEARCH INTERESTS

- Biological anthropology, osteoarchaeology, and human skeletal biology
- High-altitude adaptation, growth and development, and health in past populations
- Human body size and proportions across ecological and historical contexts
- Demography, inequality, violence, and social change in archaeological populations
- Quantitative and computational archaeology; Bayesian inference and spatiotemporal modelling
- Commingled human remains, secondary burial, and mortuary formation processes

ACADEMIC APPOINTMENTS

2026-present **Research Assistant Professor, Institute for the Humanities and Social Sciences, The University of Hong Kong**
Research in biological anthropology, osteoarchaeology, quantitative/computational archaeology, high-altitude adaptation, and demography and health in past populations.

EDUCATION

2021-2025
Cambridge, UK **PhD in Archaeology / Biological Anthropology, University of Cambridge**
Department of Archaeology. Dissertation research on high-altitude adaptation, human body size and proportions, and comparative skeletal variation in ancient Tibetan and lowland Chinese populations.

2018-2021
Chengdu, China **MA in Archaeology, Sichuan University**
School of Archaeology and Museology. GPA: 3.8/4.0. Thesis: Research on human skeletal remains from the Phiyang-Dungkar tombs in Western Tibet.

2014-2018
Chengdu, China **BA in Archaeology, Sichuan University**
School of Archaeology and Museology. GPA: 3.6/4.0.

MAJOR AWARDS AND HONOURS

2021-2025 **Harding Distinguished Postgraduate Scholars Programme, University of Cambridge**
Full PhD scholarship and research support.

2022 **International Committee of the Red Cross (ICRC) Research Scholarship**
Research scholarship supporting work related to missing persons and unidentified human remains.

2018-2021 **National Scholarship for Master's Degree, China**
First Class.

2020 **Outstanding Student Leader Award, Graduate School, Sichuan University**

2018 **Outstanding Undergraduate Dissertation, Sichuan University**

2018 **China Guardian & Xu Bangda Art Scholarship**

2017 **Outstanding Undergraduate Student Award, Sichuan University**

PUBLICATIONS

Journal articles

Cao, D., Crema, E. R., & Pomeroy, E. (2026). Multifactorial drivers of spatial variation in human body form across modern China. Scientific Reports. <https://doi.org/10.1038/s41598-026-57537-8>

- Wang, X., Yang, A., Yang, F., Yuan, H., Cao, D., Li, J., Huo, W., Lu, S., & Li, S. (2026). Subsistence change in Iron Age to Tubo-period western Xizang: A stable isotope study from Phiyang Dunkar. *Journal of Archaeological Science: Reports*, 70, 105639. <https://doi.org/10.1016/j.jasrep.2026.105639>
- Cao, D., Liu, X., Gu, W., Zhu, H., Zhang, R., Zhou, Z., Wei, Q., Zou, J., Qiu, Y., Chen, J., Ding, L., Pomeroy, E., & Yuan, H. (2025). Dental caries as indicators of agricultural practices in the foothills of Neolithic China. *Archaeological Research in Asia*, 43, 100645. <https://doi.org/10.1016/j.ara.2025.100645>
- Wang, X., Cao, D., Gu, W., Wei, Q., Chen, X.-L., Zhang, R., Ding, L., Tang, Z., Teng, X., Wang, T., Zou, J., Qiu, Y., Fu, Q., & Yuan, H. (2025). Multi-isotope analysis reveals human millet-based diets and limited mobility in the Central Plains of China ca. 5000 years ago. *International Journal of Osteoarchaeology*, 35, e3407. <https://doi.org/10.1002/oa.3407>
- Cao, D., Crema, E. R., & Pomeroy, E. (2024). Estimating intralimb proportions for commingled remains. *International Journal of Osteoarchaeology*. <https://doi.org/10.1002/oa.3326>
- Yi, B., Yuan, H., Wang, T., Cao, D., Yang, Y., & Hu, Y. (2024). Life history of a high-status noblewoman from the late Shu state during the Eastern Zhou period (770-221 BC): Childhood stress and adult stability. *International Journal of Osteoarchaeology*, 34(1).
- Tang, L., Wilkin, S., Richter, K. K., et al. (2023). Paleoproteomic evidence reveals dairying supported prehistoric occupation of the highland Tibetan Plateau. *Science Advances*, 9, eadf0345. <https://doi.org/10.1126/sciadv.adf0345>
- Tao, L., Yuan, H., Zhu, K., Liu, X., Guo, J., Min, R., He, H., Cao, D., Yang, X., Zhou, Z., Wang, R., Zhao, D., Ma, H., Chen, J., Zhao, J., Li, Y., He, Y., Suo, D., Zhang, R., Li, S., Li, L., Yang, F., Li, H., Zhang, L., Jin, L., & Wang, C.-C. (2023). Ancient genomes reveal millet farming-related demic diffusion from the Yellow River into southwest China. *Current Biology*, 33(22), 4995-5002.e7. <https://doi.org/10.1016/j.cub.2023.09.055>
- Lu, H., Li, Z., Ciren, C., Cao, D., Gao, X., & Li, L. (2022). Sding Chung: An early multi-burial cave site on the Tibetan Plateau. *Antiquity*, 96(387), 745-753. <https://doi.org/10.15184/aqy.2022.42>
- Yuan, H., Gu, W., Wei, Q., Wu, Q., Ding, L., & Cao, D. (2022). Analysis of dental caries in the Yangshao population at the Qingtai site, Zhengzhou. *Acta Anthropologica Sinica*, 41(2), 226-237. [in Chinese] <https://www.anthropol.ac.cn/EN/10.16359/j.1000-3193/AAS.2021.0023>
- Cao, D., & Yuan, H. (2021). Preliminary analysis of commingled human remains. *Yellow River, Yellow Earth, Yellow Race*, (06), 41-44. [in Chinese]

Newsletter and thesis

- Cao, D. (2017, November 17). Seeing humans through bones: A multidimensional perspective on ancient human life - Summary of the Second Annual Meeting of the Osteoarchaeology Committee of the Chinese Archaeological Society. *China Cultural Relics News*, (006). [in Chinese]
- Cao, D. (2021). Research on human skeletal remains from the Phiyang-Dungkar tombs in Western Tibet. Master's thesis, Sichuan University, Chengdu, China. [in Chinese] <https://kns.cnki.net/KCMS/detail/detail.aspx?dbname=CMFDTEMP&filename=1022445375.nh>

CONFERENCE PRESENTATIONS

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| 2025 | Cao, D., Crema, E. R., Lü, H., Yuan, H., & Pomeroy, E. Tracking spatial and temporal variation in body form across ancient Chinese populations. Annual Conference of Computer Applications and Quantitative Methods in Archaeology (CAA UK), Cambridge, United Kingdom. (Podium) |
| 2025 | Cao, D., Crema, E. R., Lü, H., Yuan, H., & Pomeroy, E. Bayesian models reveal spatiotemporal patterns of body size variation in Holocene Chinese populations. Annual Meeting of the Human Osteoarchaeology Committee, Chinese Archaeological Society, Yili, Xinjiang, China. (Podium) |
| 2025 | Cao, D., Crema, E. R., & Pomeroy, E. Tracking long-term and geographic variation in body size among Chinese populations: Combined insights from contemporary and archaeological evidence. 94th Annual Meeting of the American Association of Biological Anthropologists, Baltimore, USA. (Podium) |
| 2024 | Cao, D., Pomeroy, E., Lu, H., Yuan, H., Yang, F., & Li, S. An analysis of commingled human remains from two Bronze to Iron Age sites in West Tibet. 30th Annual Meeting of the European Association of Archaeologists, Rome, Italy. (Podium) |
| 2024 | Cao, D., Liu, X., & Yuan, H. Dental caries as indicators of agricultural practices in the foothills of Neolithic China. 24th European Meeting of the Paleopathology Association, Leiden, Netherlands. (Poster) |
| 2023 | Cao, D., Crema, E. R., & Pomeroy, E. Estimating intra-limb proportions for commingled remains. 23rd Annual Conference of the British Association for Biological Anthropology and Osteoarchaeology, London, UK. (Poster) |
| 2023 | Cao, D., Lu, H., Yang, F., Li, S., Wangdui, X., Yuan, H., & Pomeroy, E. Bioarchaeological insights into health and cultural practices on the ancient Tibetan Plateau in Central Asia. 29th Annual Meeting of the European |

- 2019 **Association of Archaeologists, Belfast, UK. (Poster)**
Cao, D., & Yuan, H. Life-course study of Zhao Xin, an official of the Northern Qi Dynasty. Third Annual Meeting of the Human Osteoarchaeology Committee, Chinese Archaeological Society, Qingdao, China. (Poster)
- 2018 **Cao, D. Dental caries in the Late Neolithic Gaoshan population in southwest China. Osteoarchaeology Session, Second Chinese Archaeological Congress, Chengdu, China. (Podium)**

FIELDWORK EXPERIENCE

Excavations

- 2021 **Sdingchung Cave Site, Zhongba, Xizang (Tibet), China**
 Participated in excavation and was responsible for analysis of human skeletal remains.
- 2016 **Haimenkou Site, Jianchuan, Dali, Yunnan, China**
 Participated in excavation and artefact sorting.

Surveys and skeletal documentation

- 2025 **Shuanghe Cave, Suiyang, Zunyi, Guizhou, China**
 Field survey.
- 2020 **Shidaqiu Site, Maerkang, Aba, Sichuan, China**
 Field survey and in situ sorting of human skeletal remains.
- 2019 **Xincun Site, Wuding, Chuxiong, Yunnan, China**
 Documentation and sorting of human skeletal remains.
- 2019 **Ladian Site, Yuanmou, Chuxiong, Yunnan, China**
 Field survey and in situ sorting of human skeletal remains.
- 2019 **Jiangxifen Site, Wuding, Chuxiong, Yunnan, China**
 Documentation and sorting of human skeletal remains.
- 2018 **Zhijiaozhongxin Site, Fenyang, Shanxi, China**
 Field survey, documentation, and collection of human skeletal remains in situ.
- 2018 **Qingtai Site, Zhengzhou, Henan, China**
 Field survey, documentation, and sorting of human skeletal remains.

TEACHING AND SUPERVISION

- 2024-2025 **Supervisor, From Data to Interpretation, Department of Archaeology, University of Cambridge**
 Coordinator: Dr Enrico R. Crema. Supervised students in data analysis tools, including R, with emphasis on archaeological and anthropological datasets.
- 2023-2024 **Supervisor, Comparative Human Biology, Department of Archaeology, University of Cambridge**
 Coordinator: Dr Mark Dyble. Supervised topics in comparative human biology, evolutionary adaptation, life history, and socio-cultural factors influencing human variability and health.
- 2023-2025 **Demonstrator, Decoding the Skeleton, Department of Archaeology, University of Cambridge**
 Coordinator: Dr Emma Pomeroy. Demonstrated skeletal analysis techniques, including osteological sorting, age and sex estimation, anthropometrics, and palaeopathological observation.
- 2024-2025 **Demonstrator, Archaeology in Action, Department of Archaeology, University of Cambridge**
 Coordinator: Dr Simon Stoddart. Demonstrated techniques in sorting and analysing human skeletal remains.
- 2018-2021 **Student Teaching Assistant, Sichuan University Bioarchaeology Laboratory, Center for Archaeological Science**
 Assisted with laboratory management, equipment maintenance, sample sorting, and teaching.

PUBLIC ENGAGEMENT AND ADDITIONAL EXPERIENCE

- 2021 **Speaker, TEDxKids@Chengdu**
 "What I Found from Buried Ancient Teeth / 一颗蛀牙来认识古蜀先民".
- 2019 **Sichuan University-Harvard University Study Camp on Ancient Shu Civilisation**
 Funded by the Fairbank Center for Chinese Studies at Harvard University and the Research

SKILLS AND SOFTWARE

- Osteological analysis: skeletal sorting, demographic reconstruction, commingled remains, age/sex estimation, anthropometrics, and palaeopathological observation.
- Statistical and computational analysis: R, tidyverse, ggplot2, brms, Bayesian modelling, data visualisation, and reproducible research workflows.
- Spatial analysis: GIS, map production, spatial data handling, and spatiotemporal modelling.
- 3D imaging and morphometrics: Artec Studio, 3D Slicer, SlicerMorph, MeshLab, ImageJ.
- Languages: Chinese (native); English (fluent).